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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL5)

Assessment Tool Weightage: 40%

Duration: 60 Mins **QUESTIONS: 3** **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I MONISH L [1972260035] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student



RUBRICS FOR CALCULATION

Criteria	Understanding of Problem & Constraints	Problem Decomposition	Application of Constraints in Solution	Logical Reasoning & Strategy	Solution Accuracy & Feasibility	Creativity & Innovation	Clarity of Presentation	Total / 10
Weightage	20%	15%	20%	15%	15%	5%	10%	
Q1								
Q2								
Q3								

Signature of the Faculty: *Total Marks Awarded:*

Annexure A

Constraint-Based Problem Solving

1. A small startup company plans to set up a computer network for 50 employees across two floors. The company has a limited budget, requires reliable connectivity, and expects moderate future expansion. The building structure restricts extensive cabling between floors. As a network engineer, propose a suitable network topology, transmission media, and devices while considering cost, scalability, and installation limitations. Justify your choices based on the given constraints.

Answer: For a 50-employee startup spread across two floors with a limited budget, restricted inter-floor cabling, and a need for moderate future scalability, a hybrid topology combining star segments on each floor with a single backbone link between floors is the most practical choice. Each floor should use a star topology built around an unmanaged or lightly managed Ethernet switch, since star topology offers easy troubleshooting, fault isolation (a single cable failure only affects one device, not the whole floor), and low-cost Cat6 UTP cabling to each workstation. The two floor switches should then be connected by a single backbone uplink, ideally a fiber-optic patch cable if the inter-floor distance exceeds 100 m or shielded Cat6 cable if within range, which minimizes the amount of cabling that must physically run between floors while still providing reliable connectivity. A basic Layer 2/3 managed switch or a small router at the backbone junction allows future expansion (VLANs, additional switches) without redesigning the network. This design is justified by the constraints given: star-per-floor keeps installation cost low and simple, the single backbone link respects the building's cabling restrictions, and the hierarchical star-of-stars structure scales easily by simply adding switches or access points on either floor as the company grows, without needing a costly mesh or full rewiring.

2. An educational institution experiences heavy network congestion during peak hours due to online classes, video streaming, and file sharing. The available bandwidth is limited, and upgrading infrastructure is not immediately possible. Analyze the situation and suggest appropriate traffic management and Quality of Service (QoS) techniques to optimize performance. Explain how your solution addresses the constraints of limited bandwidth and high demand.

Answer: To address network congestion during peak hours caused by online classes, video streaming, and file sharing without new infrastructure spending, the institution should implement traffic management and QoS techniques that prioritize and shape existing bandwidth rather than increase it. First, QoS traffic classification and prioritization (e.g., using DiffServ markings) should be configured on routers/switches so that latency-sensitive traffic such as live video classes and VoIP is placed in a high-priority queue, while less time-critical traffic like large file downloads or software updates is placed in a lower-priority queue and processed only when bandwidth is available. Second, bandwidth throttling/rate-limiting can cap the maximum bandwidth allotted to non-essential applications (e.g., recreational streaming, background file-sharing/P2P traffic) during peak hours, freeing capacity for core academic activities. Third, traffic shaping and scheduling algorithms such as Weighted Fair Queuing (WFQ) can ensure that no single user or application monopolizes the link, distributing available bandwidth more equitably across simultaneous online classes. Finally, enabling caching/local content delivery for frequently accessed learning resources reduces repeated external bandwidth consumption. Together these techniques address the constraint of limited bandwidth by ensuring the available capacity is used efficiently and fairly, protecting critical educational traffic from being crowded out by high-demand, less time-sensitive traffic, all without requiring new hardware or increased bandwidth purchase.

3. A hospital requires a highly reliable network system to support patient monitoring, medical records, and emergency services. Network downtime is unacceptable, but the hospital cannot afford a fully redundant mesh network due to cost constraints. Design a network solution that ensures high fault tolerance within budget limitations. Explain your approach, including topology selection and redundancy techniques.

Answer: For a hospital that cannot tolerate downtime but also cannot afford a fully redundant mesh network, a partial-mesh (selective redundancy) topology combined with a core-distribution-access hierarchical design offers the best balance of fault tolerance and cost. Rather than connecting every device to every other device (full mesh), critical nodes only -- such as the core switches supporting patient monitoring systems, the electronic medical records server, and the emergency department -- are interconnected with redundant links (a partial mesh among just these critical points), while less critical areas (e.g., administrative offices) use a standard star topology connected to the core through single links. This is supported by dual-homed core switches and redundant power supplies at the most failure-sensitive points, along with a spanning-tree or link-aggregation protocol so that if one path fails, traffic automatically reroutes over the backup link with minimal or no downtime. Additionally, critical devices like patient-monitoring stations can be connected via redundant NICs (dual-homing) to two separate switches, ensuring a single switch failure does not disconnect them. This approach ensures high fault tolerance exactly where it matters most (life-critical systems) while keeping overall cost manageable, since only the critical segment of the network requires redundant mesh-like cabling and hardware, and the rest of the hospital uses a conventional, lower-cost star topology -- directly addressing both the reliability requirement and the budget constraint.